

Niraj Amrutkar

 +91 9960641238  nirajamr03@gmail.com  Niraj-Amrutkar  GitHub  LeetCode

PROFESSIONAL SUMMARY

Software Engineer experienced in building **scalable backend systems, distributed data pipelines and production-grade AI applications** using **Java, Python, Spring Boot, FastAPI, Kafka and OpenSearch**. Built REST APIs, microservices, event-driven systems and high-throughput document processing pipelines handling **30M+ documents**. Experienced in **RAG, hybrid search, self-hosted embeddings, GPU acceleration and private-cloud deployments**, with strong foundations in distributed systems, system design, DSA and GenAI.

TECHNICAL SKILLS

Programming	Python, Java, C/C++, SQL
Backend	FastAPI, Spring Boot, REST APIs, Microservices, API Design
Distributed Systems	Apache Kafka, Event-Driven Architecture, Distributed Processing
AI / GenAI	LLMs, RAG, LangChain, LangGraph, MCP, Embeddings
Search	OpenSearch, Elasticsearch, Vector Search, Hybrid Search, BM25, Reranking
Cloud / DevOps	Docker, Kubernetes, Helm, Prometheus, AWS
Engineering	DSA, OOP, System Design, Design Patterns

WORK EXPERIENCE

S&P Global Ratings

July 2024 – Present

Associate Software Engineer

Hyderabad, India

- Developed production-grade **Java Spring Boot** backend services with efficient service-to-service communication using **gRPC**.
- Modernized Java applications by migrating **Java 17 to Java 21** and **Gradle to Maven**, ensuring compatibility and standardized builds.
- Designed and deployed a production-grade **RAG system** for processing **40–50 GB of enterprise data** using Python, FastAPI and OpenSearch.
- Built scalable bulk ingestion pipelines for **text extraction, chunking, embedding generation and indexing**, processing **30M+ documents** through distributed Kafka workflows.
- Hosted embedding models on dedicated servers for **private-cloud inference**, eliminating external API dependencies and keeping sensitive data within private infrastructure.
- Optimized embedding generation using **GPU acceleration, asynchronous processing and batching** to maximize GPU utilization and improve processing throughput.
- Engineered **semantic and hybrid retrieval** using OpenSearch, combining **BM25 and dense embeddings** and improving top-k retrieval precision by **15%+**.
- Designed and deployed scalable **containerized microservices** using Docker, Kubernetes and Helm for highly available production deployments.

PROJECTS

Hybrid Search RAG System

LangChain, OpenSearch, FastAPI, Ollama

- Architected an end-to-end **RAG pipeline** covering ingestion, chunking, embeddings, vector indexing, retrieval, reranking and LLM-based answer synthesis.
- Implemented **hybrid semantic + BM25 search** with cross-encoder reranking; developed FastAPI APIs and evaluated retrieval using **precision@k** to improve answer grounding.

EDUCATION

Vishwakarma Institute of Information Technology

Nov 2020 – May 2024

B.Tech in Information Technology, CGPA: 9.27/10

Pune, India

Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Probability, System Design

GATE (CS-IT) 2024 — TCS CodeVita 2023 – Top 0.14%